



Daffodil International University
Department of Computer Science and Engineering
Course Code: CSE499 Title: FYDP (Title Defense)

Proposal Approval Form

I. 1st Member information			
Student ID:	0242310005101361	Student Name:	Arnob Das
II. 2nd Member information [if there is any. Otherwise, skip this]			
Student ID:	0242310005101583	Student Name:	Md. Shazzadur Rahman Piash
Semester:	Spring 2026	Date:	
III. Proposed Title:	Transparent Predictions: Using Interpretable Dynamic Model to Understand University Student Dropout Factors		
IV. Updated Title (If there is any update):			
This is the approval for the Final Year Design Project (FYDP) proposal and title submitted by the above-mentioned student who is from the Department of CSE under my supervision. Upon thorough review and evaluation, it is my professional opinion that the proposed project title [mentioned in Sec III and IV] is both relevant and aligned with the academic goals and standards of our department. I (Supervisor) have endorsed the proposed proposal and title. Under our department's guidelines, the student has fulfilled all the necessary prerequisites and will attach this Proposal Approval Form to their FYDP Title Phase Evaluation Report.			
Supervisor Name:	Mr. Saiful Islam		
Supervisor Designation:	Assistant Professor		
Approved by:  Supervisor Signature with Date			

V. Detailed Proposal

1. Objective

This research aims to develop a transparent and dynamic prediction model that identifies university students at risk of dropping out while explaining the reasons behind each prediction. The project will build a dynamic temporal model that tracks student behaviors over time, updating risk predictions continuously. It will incorporate pre-admission records, demographics, financial status, and academic progress. Then Apply SHAP to explain why students are flagged as at-risk and reveal critical thresholds. Create actionable insights for targeted interventions.

The problem being solved is the lack of transparency in existing dropout prediction models. Current systems predict risk but fail to explain the reasons behind predictions, leaving educators unable to design effective interventions.

2. Motivation

University student dropout remains a persistent global challenge with far-reaching consequences. First-year students are particularly vulnerable, as the transition to university life presents academic, social, and financial adjustments that can overwhelm even capable individuals. In many regions, private institutions report dropout rates significantly higher than their public counterparts, with first-year students accounting for the majority of attrition.

Importance of This Research:

1. **Human Impact:** Each dropout represents unfulfilled potential, lower lifetime earnings, and reduced career opportunities.
2. **Institutional Impact:** Universities lose revenue, face reputation damage, and waste resources.
3. **Societal Impact:** Higher dropout rates reduce workforce skills and economic productivity

3. Background Study

Student dropout prediction has emerged as a significant area of research as universities seek to improve retention rates and support student success. Researchers have applied various artificial intelligence and machine learning techniques to identify at-risk students, with studies spanning multiple countries and educational contexts..

Early research used statistical methods like logistic regression for interpretability. Later, random forests gained popularity for higher accuracy. Neural networks show promise but raise interpretability concerns. Academic performance, especially first-semester grades and credits earned, consistently predicts dropout risk. Attendance patterns strongly correlate

with withdrawal decisions. Financial factors including scholarships, loans, and tuition payments also matter. Demographic characteristics provide additional predictive power. Most models provide static end-of-semester predictions, limiting intervention timing. The most accurate models function as black boxes, offering no insight into their reasoning. Threshold identification remains unexplored—research does not reveal how many absences trigger high risk. Interaction effects between factors are poorly understood.

4. Gap Analysis

Existing models achieve high accuracy but cannot explain predictions. Most models provide a single static prediction at semester end. Many ignore pre-admission data that could enable earlier predictions. Current research does not reveal critical thresholds such as how many absences matter. Predictions do not guide intervention strategies.

SHAP will provide per-student explanations of risk factors. LSTM will process temporal sequences and update risk continuously. Pre-admission data will be incorporated for earlier predictions. SHAP dependence plots will reveal critical thresholds. Personalized risk reports will explain why each student is at risk and suggest interventions.

5. Methodology

▪ Techniques/Tools:

- Python is used as the main programming language for system development.
- Pandas and NumPy are used for data cleaning, processing, and numerical manipulation.
- Matplotlib and Seaborn are used to visualize data patterns and model results.
- Scikit-learn is used for baseline machine learning models, feature selection, and data preprocessing.
- TensorFlow and Keras are used to build and train the LSTM-based dynamic prediction model.
- SHAP is used for model interpretability, feature importance analysis, and decision threshold identification.
- SMOTE is applied to handle class imbalance in the dataset.

▪ Data Collection:

These are the types of data will be collected:

- Obtain anonymized student records from partner universities with appropriate ethical approvals

- Include pre-admission data: high school GPA, entrance exam scores, high school type
- Include demographic data: age at enrollment, gender, nationality, residential status
- Include family background: parents' education and occupation, family income
- Include financial data: scholarship status, student loan status, tuition payment regularity
- Include first semester academic data: courses enrolled, courses approved, grades, credits, failed courses
- Include second semester academic data for temporal analysis
- Target variable: dropout status at end of academic year

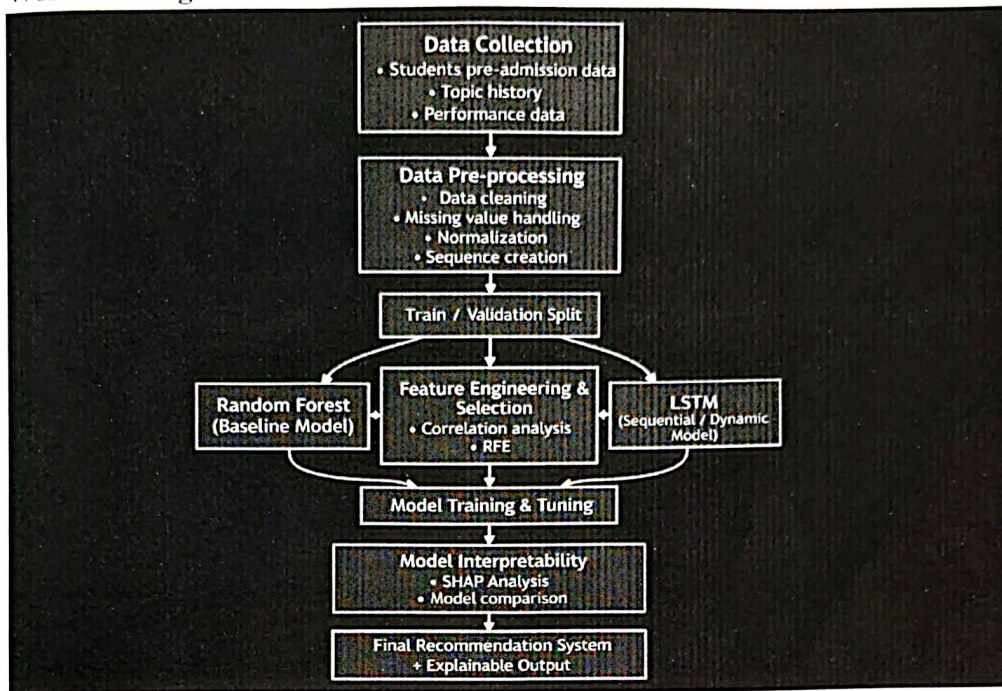
Data Analysis:

Model evaluation will employ multiple metrics. Accuracy measures overall correct predictions. Precision indicates the percentage of flagged students who actually drop out. Recall shows the proportion of actual dropouts correctly identified. F1-Score balances precision and recall for imbalanced data. ROC-AUC assesses discrimination ability. Confusion matrix provides detailed breakdown of predictions.

The experimental design includes five comparisons. Experiment one establishes baseline performance using all features. Experiment two applies feature selection to identify optimal predictors. Experiment three evaluates dynamic LSTM against static models. Experiment four quantifies the value of pre-admission data. Experiment five applies SHAP for explanations and threshold identification.

Interpretability analysis proceeds through multiple levels. Global SHAP reveals feature importance across all students. Local SHAP explains individual predictions through force plots and waterfall diagrams. Dependence analysis identifies non-linear relationships and critical thresholds where risk sharply increases. Interaction analysis explores how factors combine to influence dropout risk.

Workflow Diagram:



6. Expected Outcomes

- Evidence on whether LSTM outperforms static models for dropout prediction
- Quantification of pre-admission data value for early warning
- Discovery of critical risk thresholds
- Feature importance rankings from static and temporal models
- SHAP methodology for educational research
- Interaction effects between key factors

7. Conclusion

This research addresses a critical flaw in current dropout prediction systems: they predict but do not explain. Universities receive at-risk lists without understanding whether problems stem from attendance, academics, or finances. Without explanations, interventions become guesswork.

This project combines LSTM for temporal modeling with SHAP for interpretability, creating a transparent system that tracks evolving risk while explaining each prediction. Pre-admission data enables earlier identification, and threshold analysis reveals when intervention becomes critical.